

Technical Design – Multi-Source Candidate Data Transformation Pipeline

1. Pipeline / Step Breakdown

Detect Input → Upload Recruiter CSV & GitHub JSON → Parse → Normalize (name, email, phone, skills) → Merge using Candidate ID → Remove duplicate skills → Calculate Confidence → Generate Provenance → Validate → Export final_candidates.json → Display Flask Dashboard.

2. Canonical Output Schema & Normalization

Schema: candidate_id, full_name, emails, phones, location, current_company, headline, skills, github_name, bio, links, years_experience, experience, education, provenance, confidence.

Normalization: lowercase emails, standardized phone numbers, normalized skill names (SQL/API/HTML/AWS), duplicate skill removal, missing values shown as N/A.

3. Merge / Conflict Resolution & Confidence

Candidate ID is used for matching. Recruiter CSV is the primary source for personal information while GitHub enriches bio, GitHub details and skills. Duplicate skills are removed. Confidence is assigned according to profile completeness (email, phone, company, title, location, skills, GitHub information and bio).

4. Runtime Custom Output Configuration

The Flask application accepts CSV and JSON uploads at runtime, validates inputs, performs normalization, merging, provenance generation and confidence calculation, exports the canonical JSON output and displays the transformed profiles through the dashboard.

5. Edge Cases & Deliberately Left Out

Handled: missing email/phone/location, duplicate skills, inconsistent skill capitalization, missing GitHub information and incomplete profiles.

Left Out: LinkedIn API integration, resume PDF parsing, automatic experience extraction, AI-based ranking and database integration due to time constraints and can be considered as future improvements.

Conclusion

The proposed pipeline successfully transforms multi-source candidate information into a standardized canonical profile with normalization, conflict resolution, confidence estimation, provenance tracking and JSON generation. The modular architecture can be extended to support additional data sources in future.